

Page 1: AI Engineer Overview

AI Engineer Roadmap

- Understand Python programming
- Learn Data Structures & Algorithms
- Git, GitHub, Linux basics
- APIs and Backend fundamentals
- SQL and NoSQL databases
- Mathematics: Linear Algebra, Probability, Statistics

Page 2: Machine Learning Foundations

- Supervised vs Unsupervised Learning
- Regression and Classification
- Model Evaluation Metrics
- Scikit-Learn library
- Feature Engineering
- Model Deployment Basics

Page 3: Deep Learning

- Neural Networks
- PyTorch or TensorFlow
- CNNs for Computer Vision
- RNNs, LSTMs, Transformers
- Transfer Learning
- Model Optimization

Page 4: Generative AI & LLMs

- Transformers Architecture
- Hugging Face Ecosystem
- Prompt Engineering
- RAG (Retrieval-Augmented Generation)
- Vector Databases (FAISS, ChromaDB, Pinecone)
- Fine-Tuning and Evaluation

Page 5: AI Engineering Projects & Career

- Build End-to-End AI Applications
- LangChain / LangGraph
- FastAPI & Deployment
- Docker & Cloud Platforms
- Portfolio Projects
- Interview Preparation for AI Engineer Roles
- System Design for AI Applications